# DQH-BAN: Dual-Stream Quantum Healthcare Intelligence Architecture (DQH-BAN-EXP-2026-V2)

**Abstract:** We present DQH-BAN, a dual-stream quantum healthcare intelligence architecture integrating quantum-assisted cybersecurity (Stream A) and physiological signal intelligence (Stream B) anchored by a Hyperledger Fabric security orchestration layer. All metrics reported in this paper are generated directly from verified offline checkpoints across seven physical benchmarks.

| Dataset | Model | Acc (%) | Recall (%) | Macro-F1 (%) | FPR (%) | FNR (%) |
|---|---|---|---|---|---|---|
| CICIoMT2024 | XGBoost | 96.24 | 96.06 | 85.71 | 0.57 | 3.94 |
| WUSTL-EHMS-2020 | XGBoost | 97.43 | 87.62 | 94.03 | 1.17 | 12.38 |
| WUSTL-HDRL-2024 | XGBoost | 99.81 | 99.55 | 99.39 | 0.17 | 0.45 |
| CIC-IoT-2023 | XGBoost | 99.01 | 98.65 | 99.00 | 0.52 | 1.35 |
| ToN-IoT | XGBoost | 99.47 | 99.27 | 99.47 | 0.34 | 0.73 |
| CICIoMT2024 | classical_matched | 96.24 | 96.75 | 84.38 | 13.00 | 3.25 |
| CICIoMT2024 | Quantum_4q_l2_ring | 93.81 | 93.78 | 79.29 | 5.62 | 6.22 |
| WUSTL-EHMS-2020 | Quantum_4q_l2_ring | 93.38 | 52.44 | 81.43 | 0.75 | 47.56 |
| WUSTL-HDRL-2024 | Quantum_4q_l2_ring | 99.45 | 98.44 | 98.26 | 0.46 | 1.56 |
| CIC-IoT-2023 | Quantum_4q_l2_ring | 98.34 | 98.34 | 85.94 | 1.69 | 1.66 |
| ToN-IoT | Quantum_4q_l2_ring | 99.31 | 58.57 | 84.57 | 0.14 | 41.43 |

**Verification Certification:** Passed Round 1 (16 Technical Checks) and Round 2 (End-to-End Clean Session Recalculation).